

Amit_Lab2 - Day 2 Assignment

C Programming - Lab 2

Question 1: Check if a number is positive or negative

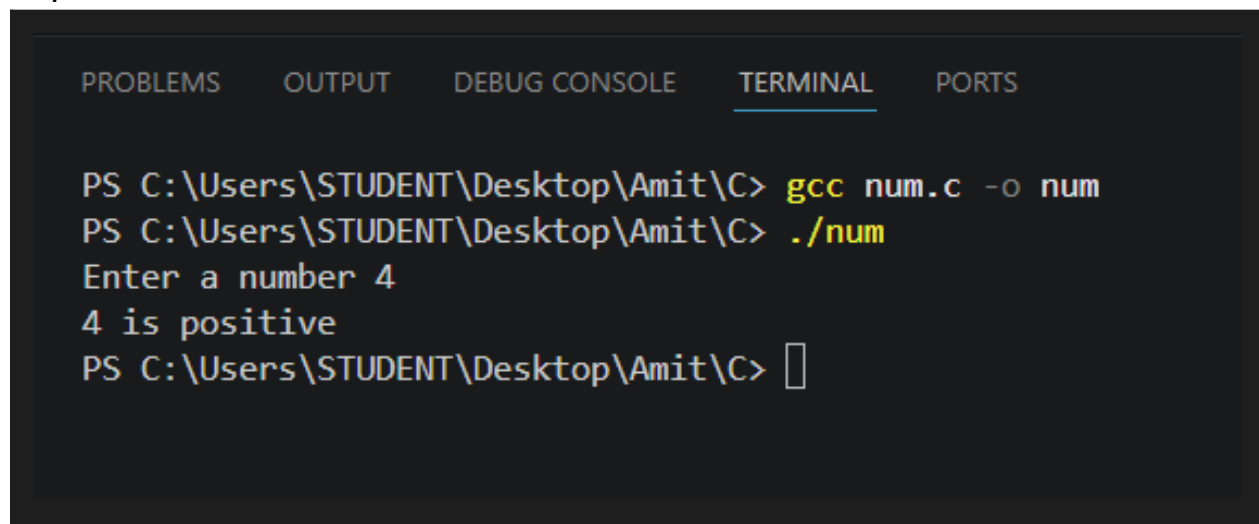
```
#include<stdio.h>

int main()
{
    int num;
    printf("Enter a number ");
    scanf("%d", &num);

    if(num>0)
    {
        printf("%d is positive\n", num);
    }
    else if(num==0)
    {
        printf("%d is zero\n", num);
    }
    else{
        printf("%d is negative\n", num);
    }

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\STUDENT\Desktop\Amit\C> gcc num.c -o num
PS C:\Users\STUDENT\Desktop\Amit\C> ./num
Enter a number 4
4 is positive
PS C:\Users\STUDENT\Desktop\Amit\C> 
```

Question 2: Check if a number is even or odd

```
#include <stdio.h>

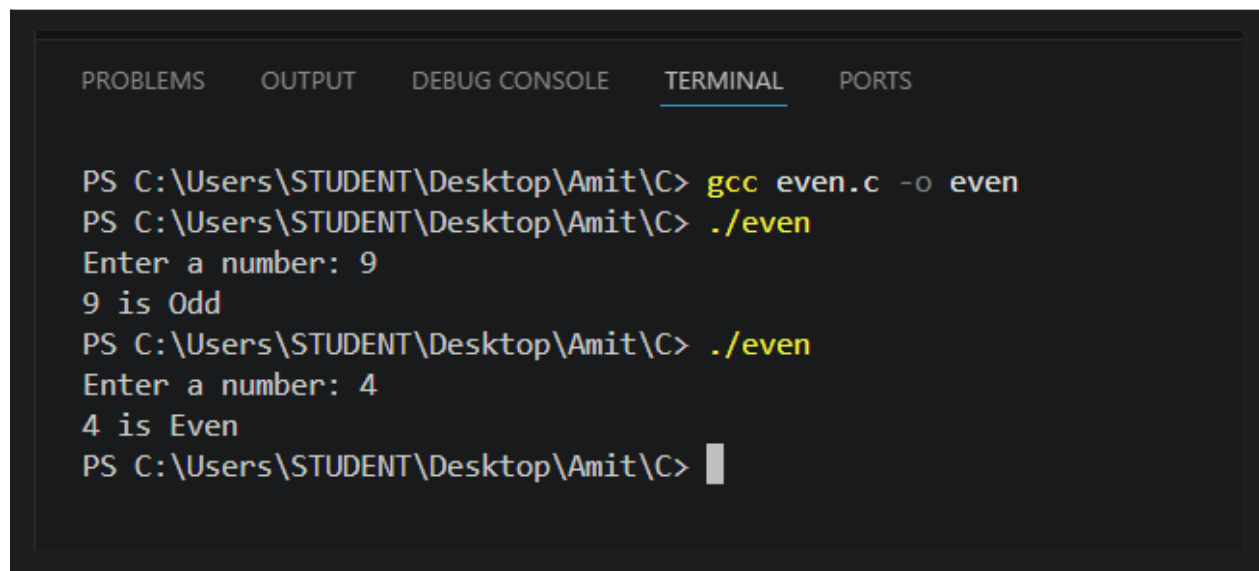
int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(num % 2 == 0)
    {
        printf("%d is Even\n", num);
    }
    else
    {
        printf("%d is Odd\n", num);
    }

    return 0;
}
```

Output:



The screenshot shows a terminal window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active. The commands and output are as follows:

```
PS C:\Users\STUDENT\Desktop\Amit\C> gcc even.c -o even
PS C:\Users\STUDENT\Desktop\Amit\C> ./even
Enter a number: 9
9 is Odd
PS C:\Users\STUDENT\Desktop\Amit\C> ./even
Enter a number: 4
4 is Even
PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 3: Check if a person is an adult or minor (18 years)

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(num>18)
    {
        printf("%d is Adult\n", num);
    }
    else
    {
        printf("%d is Minor\n", num);
    }

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\STUDENT\Desktop\Amit\C> gcc vote.c -o vote
PS C:\Users\STUDENT\Desktop\Amit\C> ./vote
Enter a number: 19
19 is Adult
PS C:\Users\STUDENT\Desktop\Amit\C> ./vote
Enter a number: 17
17 is Minor
PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 4: Check if temperature is hot or cold (30°C or above is Hot)

```
#include <stdio.h>
```

```
int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(num>30)
    {
        printf("%d is hot\n", num);
    }
    else
    {
        printf("%d is cold\n", num);
    }

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\STUDENT\Desktop\Amit\C> gcc hot.c -o hot
PS C:\Users\STUDENT\Desktop\Amit\C> ./hot
Enter a number: 32
32 is hot
PS C:\Users\STUDENT\Desktop\Amit\C> ./hot
Enter a number: 25
25 is cold
PS C:\Users\STUDENT\Desktop\Amit\C> [ ]
```

Question 5: Check if a number is divisible by 5

```
#include <stdio.h>

int main()
{
```

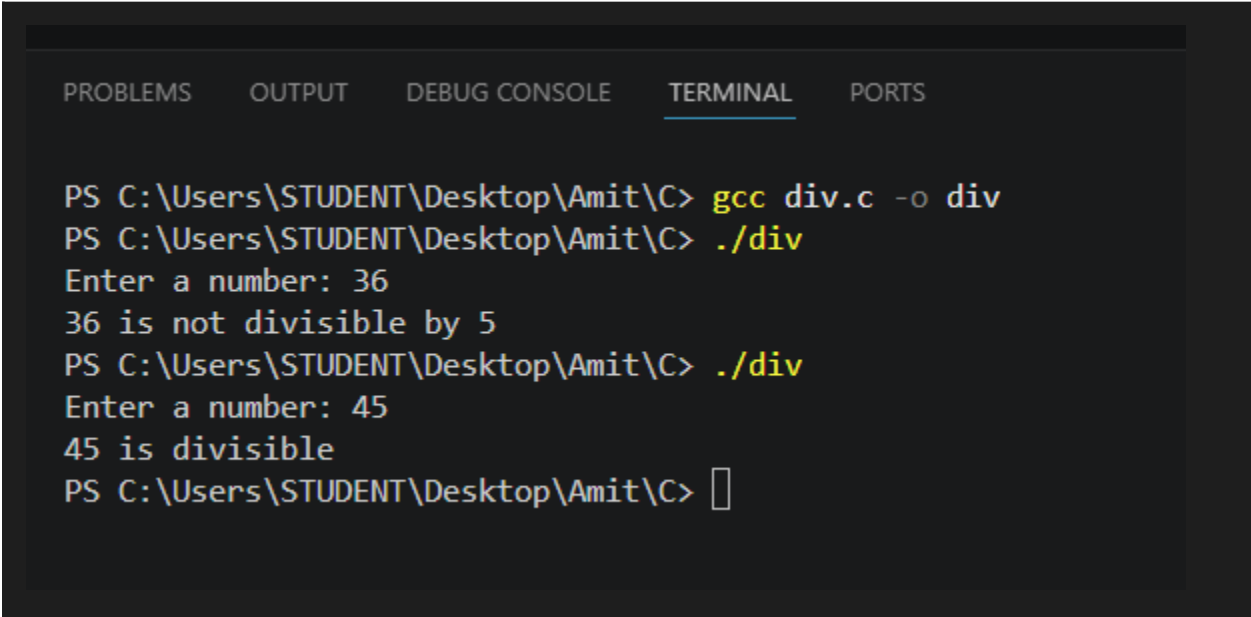
```
int num;

printf("Enter a number: ");
scanf("%d", &num);

if(num % 5 == 0)
{
    printf("%d is divisible\n",num);
}
else
{
    printf("%d is not divisible by 5\n",num);
}

return 0;
}
```

Output:



The screenshot shows a code editor with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, displaying the following commands and output:

```
PS C:\Users\STUDENT\Desktop\Amit\C> gcc div.c -o div
PS C:\Users\STUDENT\Desktop\Amit\C> ./div
Enter a number: 36
36 is not divisible by 5
PS C:\Users\STUDENT\Desktop\Amit\C> ./div
Enter a number: 45
45 is divisible
PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 6: Maximum of three numbers using Nested-if

```
#include <stdio.h>

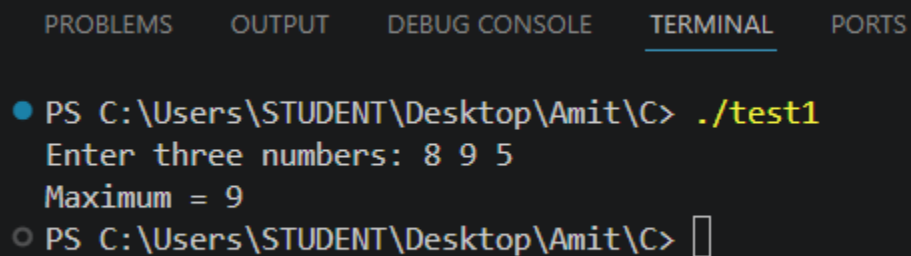
int main()
{
    int a, b, c;
```

```
printf("Enter three numbers: ");
scanf("%d %d %d", &a, &b, &c);

if(a > b)
{
    if(a > c)
    {
        printf("Maximum = %d\n", a);
    }
    else
    {
        printf("Maximum = %d\n", c);
    }
}
else
{
    if(b > c)
    {
        printf("Maximum = %d\n", b);
    }
    else
    {
        printf("Maximum = %d\n", c);
    }
}

return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\STUDENT\Desktop\Amit\C> ./test1
Enter three numbers: 8 9 5
Maximum = 9
○ PS C:\Users\STUDENT\Desktop\Amit\C> 
```

Question 7: Leap Year using Nested-if

```
#include <stdio.h>

int main()
{
    int year;

    printf("Enter year: ");
    scanf("%d", &year);

    if((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
    {
        printf("%d is a leap year\n", year);
    }
    else
    {
        printf("%d is not a leap year\n", year);
    }

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\STUDENT\Desktop\Amit\C> gcc leaps.c -o leaps
PS C:\Users\STUDENT\Desktop\Amit\C> ./leaps
Enter year: 2024
2024 is a leap year
PS C:\Users\STUDENT\Desktop\Amit\C> 
```

Question 8: Calculator using switch-case

```
/*
 * Question: Take two numbers and an operator (+, -, *, /) from the user and
 *           perform the operation using a switch-case statement.
 */
```

```
#include <stdio.h>

int main()
{
    int a, b;
    char op;

    printf("Enter first number: ");
    scanf("%d", &a);

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);

    printf("Enter second number: ");
    scanf("%d", &b);

    switch(op)
    {
        case '+':
            printf("Result = %d\n", a+b);
            break;

        case '-':
            printf("Result = %d\n", a-b);
            break;

        case '*':
            printf("Result = %d\n", a*b);
            break;

        case '/':
            printf("Result = %d\n", a/b);
            break;

        default:
            printf("Enter a valid operator\n");
    }

    return 0;
}
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\STUDENT\Desktop\Amit\C> ./operators
Enter first number: 8
Enter operator (+, -, *, /): -
Enter second number: 7
Result = 1
PS C:\Users\STUDENT\Desktop\Amit\C> |
```

Question 9: Vowel or Consonant

```
/*
 * Question: Check whether a given character is a vowel or a consonant.
 */
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    switch(ch)
    {
        case 'a':
            printf("%c is a vowel\n", ch);
            break;
        case 'e':
            printf("%c is a vowel\n", ch);
            break;
        case 'i':
            printf("%c is a vowel\n", ch);
            break;
        case 'o':
            printf("%c is a vowel\n", ch);
            break;
        case 'u':
            printf("%c is a vowel\n", ch);
            break;
    }
```

```

        default:
            printf("%c is a consonant\n", ch);
    }

    return 0;
}

```

Output:

```

• PS C:\Users\STUDENT\Desktop\Amit> cd C
• PS C:\Users\STUDENT\Desktop\Amit\C> gcc vowel.c -o vowel
• PS C:\Users\STUDENT\Desktop\Amit\C> ./vowel
Enter a character: a
a is a vowel
• PS C:\Users\STUDENT\Desktop\Amit\C> ./vowel
Enter a character: b
b is a consonant
• PS C:\Users\STUDENT\Desktop\Amit\C> ./vowel
Enter a character: c
c is a consonant

```

Question 10: Print numbers from 1 to 10

```

/*
 * Question: Print numbers from 1 to 10.
 */
#include <stdio.h>

int main()
{
    int i;

    for(i = 1; i <= 10; i++)
    {
        printf("%d ", i);
    }
    printf("\n");

    return 0;
}

```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\STUDENT\Desktop\Amit\C> gcc test2.c -o test2
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test2
  1 2 3 4 5 6 7 8 9 10
○ PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 11: Multiplication table

```
/*
 * Question: Print the multiplication table of a number.
 */
#include <stdio.h>

int main()
{
    int i, num;

    printf("Enter the number: ");
    scanf("%d", &num);

    for(i = 1; i <= 10; i++)
    {
        printf("%d x %d = %d\n", num, i, num * i);
    }

    return 0;
}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc loop.c -o loop
● PS C:\Users\STUDENT\Desktop\Amit\C> ./loop
Enter the number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
○ PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 12: Print even numbers between 1 and 50

```
#include <stdio.h>

int main()
{
    int i;

    for(i = 2; i <= 50; i = i + 2)
    {
        printf("%d\n", i);
    }

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\STUDENT\Desktop\Amit\C> ./even
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
○ PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 13: Numbers divisible by both 3 and 5 (1 to 100)

```
/*
 * Question: Print only numbers divisible by both 3 and 5 between 1 and 100.
 */
#include <stdio.h>

int main()
{
    int i;

    for(i = 1; i <= 100; i++)
    {
        if(i % 3 == 0 && i % 5 == 0)
        {
            printf("%d\n", i);
        }
    }

    return 0;
}
```

Output:

```
• PS C:\Users\STUDENT\Desktop\Amit\C> gcc numbers.c -o numbers
• PS C:\Users\STUDENT\Desktop\Amit\C> ./numbers
15
30
45
60
75
90
Focus folder in explorer (ctrl + click)
• PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 14: Sum only odd numbers up to N

```
/*
 * Question: Sum only odd numbers up to N.
 */
#include <stdio.h>

int main()
{
    int i;
    int N, sum = 0;

    printf("Enter the number: ");
    scanf("%d", &N);

    for(i = 1; i <= N; i++)
    {
        if(i % 2 != 0)
        {
            sum = sum + i;
        }
    }

    printf("Sum = %d\n", sum);

    return 0;
}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc sum.c -o sum
● PS C:\Users\STUDENT\Desktop\Amit\C> ./sum
Enter the number: 10
Sum = 25
```

Question 15: Print alphabets from a to z using for loop

```
/*
 * Question: Print alphabets from a to z using a for loop.
 */
#include <stdio.h>

int main()
{
    char ch;

    for(ch = 'a'; ch <= 'z'; ch++)
    {
        printf("%c ", ch);
    }

    return 0;
}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc alpha.c -o alpha
● PS C:\Users\STUDENT\Desktop\Amit\C> ./alpha
a b c d e f g h i j k l m n o p q r s t u v w x y z
○ PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Question 16: Print 5 stars in a row using for loop

```
/*
 * Question: Print 5 stars (*) in a row using a for loop.
 */
#include <stdio.h>

int main()
```

```

{
    int i;
    char ch = '*';

    for(i=1; i<=5; i++)
    {
        printf("%c", ch);
    }

    return 0;
}

```

Output:

```

● PS C:\Users\STUDENT\Desktop\Amit\C> ./star
*****
○ PS C:\Users\STUDENT\Desktop\Amit\C> 

```

Question 17: Square star pattern of N rows

```

/*
 * Question: Print a square star (*) pattern of N rows using a for loop.
 */
#include <stdio.h>

int main()
{
    int i,j,n;
    char ch = '*';
    printf("Enter no. of rows: ");
    scanf("%d", &n);
    for(i=1; i<=n; i++)
    {
        for(j=1; j<=n; j++)
        {
            printf("%c", ch);
        }
        printf("\n");
    }
}

```



```
}

return 0;

}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc square.c -o square
● PS C:\Users\STUDENT\Desktop\Amit\C> ./square
Enter no. of rows: 4
****
****
****
****
```

Question 18: Right triangle star pattern

```
/*
 * Question: Print a right triangle star pattern series using a for loop.
 */
#include <stdio.h>

int main()
{
    int n, i, j;
    char ch = '*';

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%c", ch);
        }

        printf("\n");
    }

    return 0;
}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc triangle.c -o triangle
● PS C:\Users\STUDENT\Desktop\Amit\C> ./triangle
Enter number of rows: 5
*
**
***
****
*****
```

Question 19: Print stars in a row using while loop (user input)

```
/*
 * Question: Print stars (*) in a row by getting user input using a while loop.
 */
#include <stdio.h>

int main()
{
    int n, i = 1;
    char ch = '*';

    printf("Enter number of stars: ");
    scanf("%d", &n);

    while(i<n)
    {
        printf("%c", ch);
        i++;
    }

    return 0;
}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc sqwhile.c -o sqwhile
● PS C:\Users\STUDENT\Desktop\Amit\C> ./sqwhile
Enter number of stars: 5
*****
```

Question 20: Print stars in a row using do while loop (user input)

```
/*
 * Question: Print stars (*) in a row by getting user input using a do while loop.
 */
#include <stdio.h>

int main()
{
    int n, i = 1;
    char ch = '*';

    printf("Enter number of stars: ");
    scanf("%d", &n);

    do
    {
        printf("%c", ch);
        i++;
    }
    while(i<=n);

    return 0;
}
```

Output:

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc sqwhile.c -o sqwhile
● PS C:\Users\STUDENT\Desktop\Amit\C> ./sqwhile
Enter number of stars: 5
*****
```

Question Bank: Answers

Q	Answer	Explanation
1	A. Greater than 5	10 > 5 is true
2	A. 5 4 3 2 1	i decrements from 5 to 1
3	B. 0 1 2 4	continue at i==3 skips printing 3
4	B. 0 1 2	break at i==3 exits the loop
5	C. 1 2 3	goto loops until i becomes 4

Q	Answer	Explanation
6	b. Two	first else if matches; second else if(x==2) is never reached
7	B. 0 1	after i==2, flag stays 1, so continue skips 2, 3, 4
8	B. 1 2 4 5	post-increment count++; continue at count==3

Question Bank: Grade Program

```
#include<stdio.h>
int main()
{
    int marks;
    printf("Enter marks: ");
    scanf("%d", &marks);

    if(marks>=90)
    {
        printf("Grade is : A");
    }
    else if(marks>=75){
        printf("Grade is : B");
    }

    else if(marks>=50){
        printf("Grade is : C");
    }

    else if(marks>=35){
        printf("Grade is : D");
    }

    else{
        printf("Fail");
    }

    return 0;
}
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc test3.c -o test3
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test3
Enter marks: 60
Grade is : C
○ PS C:\Users\STUDENT\Desktop\Amit\C> █
```

Bonus: Simple ATM Machine (if...else only)

```
/*
 * Question: Simulate a simple ATM machine using only if...else statements.
 * - Ask for a 4-digit PIN (correct PIN: 1234) -> show menu.
 * - Wrong PIN -> "Access Denied" and end.
 * - Menu: 1. Check Balance  2. Withdraw Money  3. Deposit Money
 * - Any other choice -> "Invalid choice".
 * - Print a thank-you message before ending.
 */
#include<stdio.h>
int main()
{
    int pin=1234;
    int currentbalance=5000;
    int choice, withdraw, balance, deposit, newbalance;
    printf("Enter your pin: ");
    scanf("%d", &pin);

    if(pin==1234)
    {
        printf("1. Check Balance\n");
        printf("2. Withdraw Money\n");
        printf("3. Deposit Money\n");
        printf("Select your choice: ");
        scanf("%d", &choice);
        if(choice==1)
        {
            printf("Your current balance is: %d", currentbalance);
        }
        else if(choice==2)
        {

```

```

        printf("Enter the amount you want to withdraw :");
        scanf("%d", &withdraw);
        if(withdraw<=currentbalance)
        {
            balance = currentbalance - withdraw;
            printf("Your withdrawl is successful\n ");
            printf("Your available balance is %d", balance);
        }
        else{
            printf("Available is not enough");
        }
    }
    else if(choice==3)
    {
        printf("How much you want to deposit: ");
        scanf("%d", &deposit);
        newbalance = currentbalance + deposit;
        printf("Your available balance is:%d",newbalance);
    }
    else{
        printf("Invalid choice");
    }

}
else{
    printf("Access Denied");
}

return 0;
}

```

Output (withdraw):

```

● PS C:\Users\STUDENT\Desktop\Amit\C> gcc test4.c -o test4
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test4
Enter your pin: 1234
1. Check Balance
2. Withdraw Money
3. Deposit Money
Select your choice: 3
How much you want to deposit: 200
Your available balance is:
● PS C:\Users\STUDENT\Desktop\Amit\C> gcc test4.c -o test4
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test4
Enter your pin: 1234
1. Check Balance
2. Withdraw Money
3. Deposit Money
Select your choice: 3
How much you want to deposit: 200
Your available balance is:5200
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test4
Enter your pin: 1234
1. Check Balance
2. Withdraw Money
3. Deposit Money
Select your choice: 4
Invalid choice
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test4
Enter your pin: 123456
Access Denied
● PS C:\Users\STUDENT\Desktop\Amit\C> ./test4
Enter your pin: 1234
1. Check Balance
2. Withdraw Money
3. Deposit Money
Select your choice: 2
Enter the amount you want to withdraw :7000
Available is not enough
○ PS C:\Users\STUDENT\Desktop\Amit\C> 

```

Conceptual Questions

1. Which loop should be used when the number of iterations is known in advance? Why?

The for loop. Because when the number of iterations is known, everything can be written in one line: the initialization, the condition, and the increment, e.g. for(i = 1; i <= 10; i++).

2. Which loop is suitable when the number of iterations is not known in advance? Explain with an example.

The while loop. It checks the condition before running the body, so it is used when the loop depends on conditions that change at runtime. Example: keep reading numbers until the user enters -1.

```

int num;
printf("Enter numbers (-1 to stop): ");

```

```
scanf("%d", &num);
while(num != -1)
{
    printf("%d\n", num);
    scanf("%d", &num);
}
```

3. Which loop guarantees that its body executes at least once? Why?

The do-while loop. Because it checks the condition at the end, after the body has already run once.

4. What is the difference between entry-controlled and exit-controlled loops?

Entry-controlled loops (for, while) test the condition before entering the body, so the body may execute zero times. Exit-controlled loops (do-while) test the condition after the body, so the body executes at least once.

5. Why is a semicolon required after the condition in a do-while loop?

The semicolon terminates the while(condition); statement. It is part of the do-while syntax that tells the compiler where the loop ends.

6. What is the difference between break and continue in C?

break immediately exits the loop (or switch) entirely and continues after it. continue skips the rest of the current iteration and jumps to the next iteration of the loop.

7. Can a switch statement completely replace an if-else statement in every situation? Explain.

No. switch can only compare a value for equality against constant integer/character values. It cannot handle ranges, relational conditions, floating-point values, or compound boolean conditions. if-else is more flexible and can replace switch, but not the other way around.

8. What types of values can generally be used with case labels?

Case labels must be constant expressions of integer type: int, char, or enum values. Floating-point values, variables, and strings are not allowed.

9. What is the purpose of the sizeof operator in C? Does sizeof return the size in bits or bytes?

sizeof returns the memory size of a type or variable in bytes (not bits). Example: sizeof(int) usually returns 4 on most systems.

10. How many times will the inner statement execute?

12 times. The outer loop runs 3 times and the inner loop runs 4 times each time: $3 \times 4 = 12$.

```
for(i = 1; i <= 3; i++)
{
    for(j = 1; j <= 4; j++)
    {
        printf("*");
    }
}
```